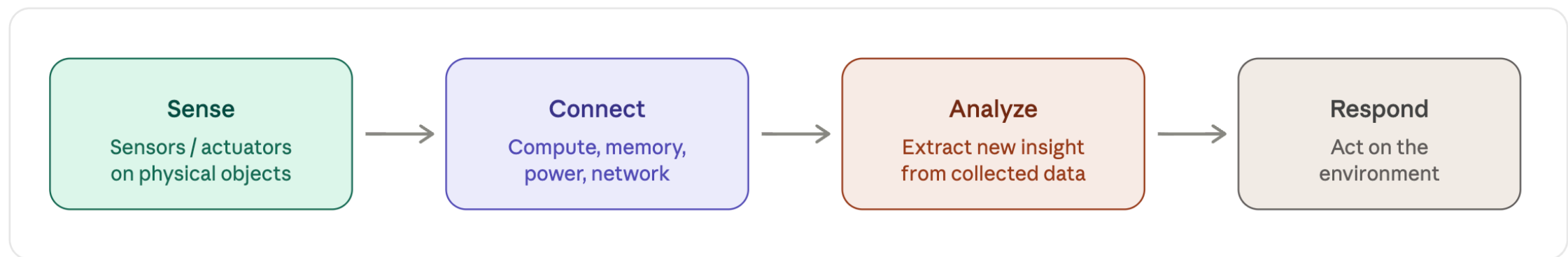


!"#\$%&'%()*+

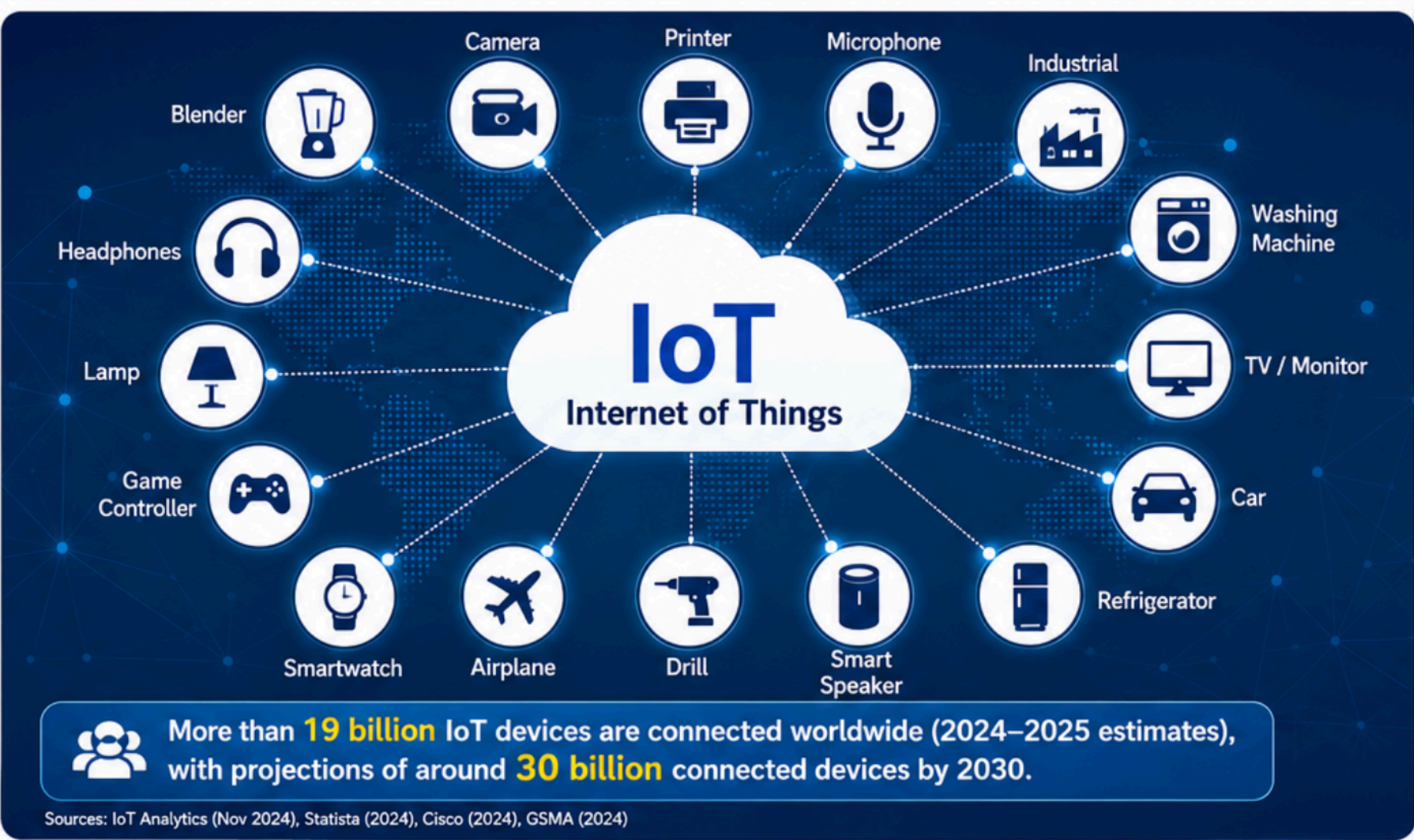
- Internet of Things (IoT) is the **network of smart physical objects**
 - Physical objects (e.g. devices, vehicles, buildings, etc.) **embedded with** sensors/actuators, computation unit, memory unit, power source, and network connectivity,
 - **Which** enables the physical object to **collect and exchange data**,
 - **Analyze** the collected data to extract new insight and **respond** accordingly.



- Goal of IoT is to “**connect the unconnected**”
 - “Things” or “objects” **that were not supposed to be connected** to the Internet

IoT: Technology Transition in Traditional Computer Networks

From connected devices to an intelligent, unified, and scalable digital ecosystem



Key Improvements with IoT



Seamless Connectivity

IoT connects devices, machines, and tools using wireless and wired technologies, anytime and anywhere.



Massive Scale

More than **19 billion** IoT devices are connected worldwide (2024–2025), with projections of around **30 billion** by 2030.



Global Reach

IoT enables global communication, allowing real-time monitoring, remote access, and centralized control.



Technology Unification

Integration of low-power embedded systems, cloud computing, big data, AI/ML, and advanced networking.



Intelligence & Automation

IoT enables data-driven insights, smart decision-making, predictive analytics, and autonomous operations.



Impact

- ✓ Higher efficiency and productivity
- ✓ Better decision-making
- ✓ Cost optimization
- ✓ Enhanced user experience
- ✓ Enabling smart cities and industries
- ✓ Sustainable and scalable future

Technology Transition Journey



1. Traditional Networks

Standalone systems with limited connectivity.



2. Connected Devices

Basic internet connectivity across devices.



3. IoT Adoption

Multiple devices connected and data exchange begins.



4. Intelligent Systems

AI, analytics, and automation drive smart operations and predictive capabilities.



5. Unified Ecosystem

Interconnected ecosystem with real-time intelligence and seamless collaboration.



IoT is transforming how we connect, communicate, and collaborate—building a smarter, safer, and more sustainable future.

Cont...

- Unifications of technologies:
 - Embedded systems,
 - Low power and low rate network,
 - Internet,
 - Big data,
 - Data analytics,
 - Cloud computing,
 - Edge Intelligence
 - Software defined networks,
 - Network and data security
 - Etc.

Alternate Definition:

“The Internet of Things (IoT) is the **network of physical objects** that contain embedded technology to **communicate** and **sense** or **interact** with their internal states or the external environment.”
‘Gartner Research*

Brief History of IoT

The term "[Internet of things](#)" was likely coined by [Kevin Ashton](#) of Procter & Gamble, later MIT's Auto-ID Center, in 1999.

- “In the 20th century, [computers were brains without senses](#)‘they only knew what we told them.” Now in the 21st century, [computers are sensing things for themselves](#)! ‘Kevin Ashton
- Early 1980 sat the Carnegie Mellon University, a group of students created a way to get their campus [Coca-Cola vending machine](#) to report on its contents through a network in order to save them the trek if the machine was out of Coke.
- In 1990, John Romkey, developer of the first TCP/IP stack for IBM PC in 1983, connected a [toaster](#) to the internet for the first time.
- In 1991, a group of students at the University of Cambridge used a web camera to report on coffee available in their computer labs [coffee pot](#).
- At the beginning of the 21st Century, LG Electronics introduced the world’s first [refrigerator](#) connected to the internet

Growth of Intelligence

- The popularity of the term IoT did not accelerate until **2010/2011** and reached mass market in **2013–14**.
- Definition of the IoT has evolved over time and continues to expand with emerging technologies.



IoT in 2026 and Beyond
19.8+ billion connected IoT devices in 2025 and projected to reach **~34.2 billion** by 2030.
Source: Statista, IDC (2025)

